

FINANCE RAG — SYSTEM ARCHITECTURE & PROJECT REPORT

Enterprise Multi-Quarter Financial Document Intelligence & Grounded Question-Answering

Target Entity: Tesla, Inc. (TSLA) FY 2023 (Q1, Q2, Q3, Q4) | **LLM:** Meta Llama-3.1-70B (NVIDIA NIM) | **Embeddings:** nv-embedqa-e5-v5 | **Vector DB:** ChromaDB

1. Executive Summary & Problem Motivation (The "Why")

Quarterly corporate earnings reports (10-Q disclosures and shareholder updates) are rich in mission-critical data but highly challenging to analyze rapidly. A single fiscal year comprises hundreds of pages of condensed financial tables, footnotes, operational metrics, and management commentary. Investment analysts and corporate researchers spend substantial manual effort extracting cross-quarter trends, verifying margin trajectories, and assessing risks.

Core Objectives of the Finance RAG System:

- **Zero Hallucination Tolerance:** Financial research desks require 100% factual fidelity. The system enforces strict prompt guardrails requiring honest refusals for unavailable or out-of-scope data.
- **Multi-Quarter Context Synthesis:** Enables comparative questions across all 4 fiscal quarters (e.g., net profit highs, margin trend trajectory, YoY revenue shifts).
- **Auditable Source Grounding:** Every generated response is linked directly to exact document filenames and page numbers for compliance and instantaneous verification.
- **Dual Interface Delivery:** Interactive Streamlit UI with token-level streaming for human analysts alongside a high-throughput FastAPI REST API for enterprise backend automation.

2. System Architecture & Engineering Stack

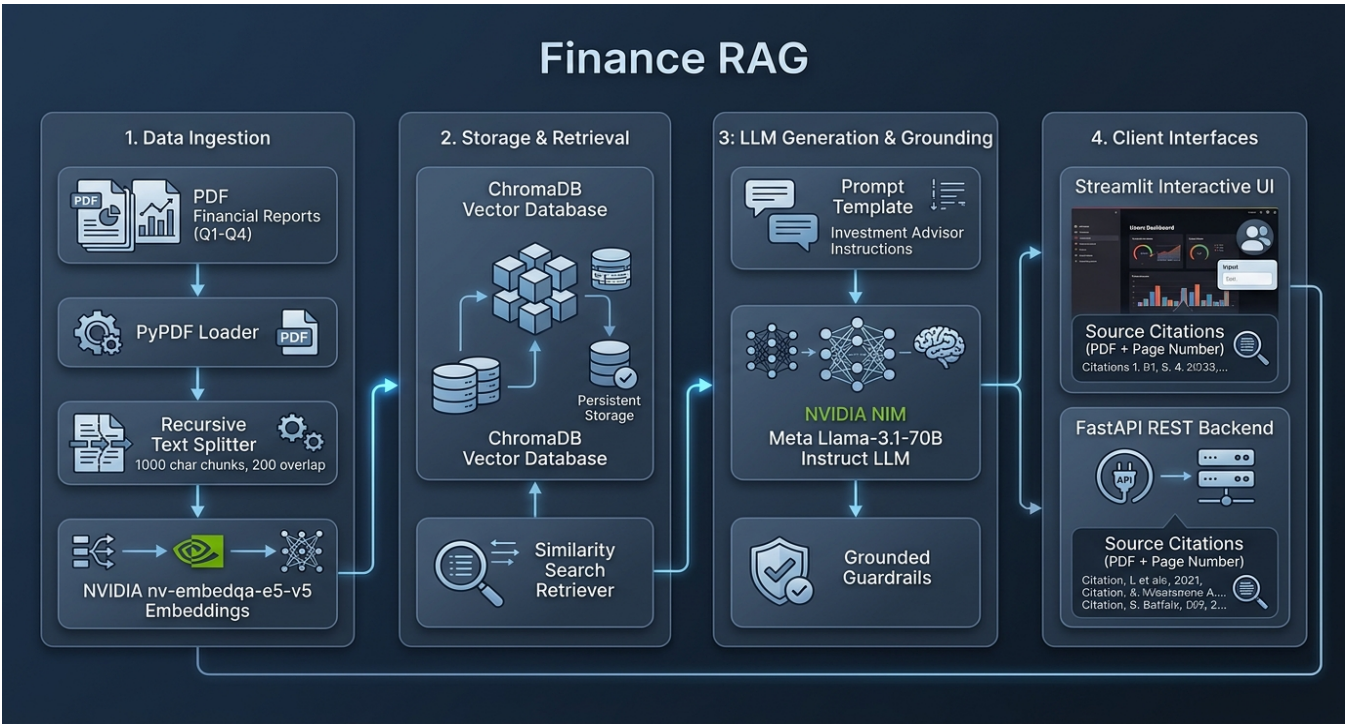


Figure 1: End-to-End Finance RAG Pipeline — Ingestion, ChromaDB Vector Store, NVIDIA NIM Guardrailed LLM, and Dual Interfaces.

3. Technical Pipeline Breakdown & Design Decisions

Pipeline Stage	Component / Configuration	Engineering Rationale & Details
Document Ingestion	PyPDF Loader langchain_community	Extracts textual contents, table layouts, and preserves original 0-indexed page numbers mapped to PDF metadata.
Chunking Strategy	Size: 1000 characters Overlap: 200 characters	1000 characters encapsulates complete financial statement rows and paragraphs without context truncation; 200-char overlap prevents boundary sentence splitting.
Embedding Model	nvidia/nv-embedqa-e5-v5 (NVIDIA NIM)	State-of-the-art 4096-dimension dense retrieval model optimized specifically for QA retrieval and tabular/financial semantic similarity.
Vector Storage	ChromaDB (Local Disk Persistence)	High-performance local vector store with SQLite metadata indexing, facilitating fast similarity retrieval without external database overhead.
Generation Model	meta/llama-3.1-70b-instruct (NVIDIA NIM, temp=0.1)	70-billion parameter reasoning LLM hosted on high-throughput NVIDIA NIM endpoints. Low temperature (0.1) minimizes hallucinations.
Guardrail Persona	Investment Advisory Research Desk Persona	Explicit directive to answer ONLY from context; mandating exact refusal string: <i>'The information is not available in the uploaded documents.'</i>

4. Application Interfaces & User Experience

The project provides two production-grade interfaces designed for both human analysts and automated pipelines:

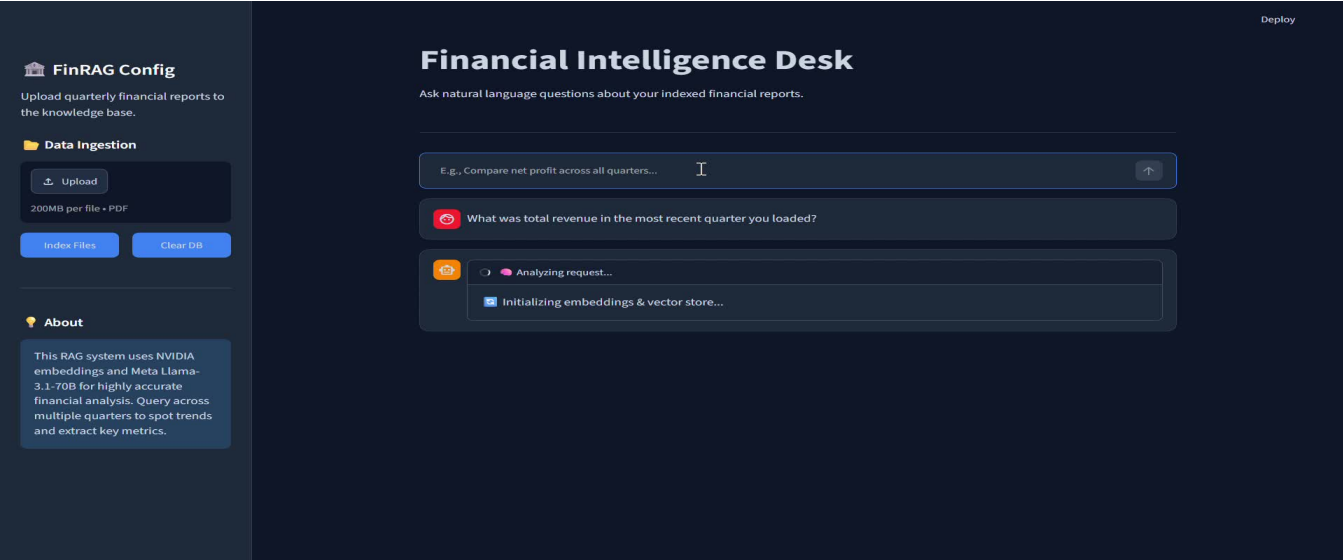


Figure 2: Streamlit Interactive Desk with real-time status tracker, streaming answers, and expandable source badges.

- **Streamlit Intelligence Desk:** Features drag-and-drop PDF ingestion, live indexing progress, real-time chunk statistics, token-by-token streaming responses (`st.write_stream()`), and expandable source citation badges displaying exact PDF filenames and page numbers.
- **FastAPI REST Backend:** Exposes interactive Swagger documentation at `/docs` with endpoints: `POST /ingest` (multipart document indexing), `POST /ask` (natural language question querying), and `GET /stats` (collection chunk metrics).

5. Comprehensive Test Questions & Benchmark Results

The Finance RAG system was evaluated across 10 rigorous financial, comparative, trend, and adversarial questions on the loaded Tesla (TSLA) FY 2023 quarterly reports (Q1–Q4). Below is the complete record of verified outputs and source citations:

#	Question & Intent	RAG System Answer	Verified Sources
Q1	Total Revenue (Latest Quarter) <i>Factual retrieval</i>	The most recent quarter loaded is Q4 2023. Total revenue in Q4 2023 was \$25,167 million (\$25.2B) .	TSLA-Q4-2023.pdf (p. 4, 6) TSLA-Q2-2023.pdf (p. 5)
Q2	Net Profit Across Quarters <i>Cross-quarter comparison</i>	• Q1 2023: \$2,513M GAAP / \$2,931M Non-GAAP • Q2 2023: \$2,703M GAAP / \$3,148M Non-GAAP • Q3 2023: \$1,853M GAAP / \$2,318M Non-GAAP • Q4 2023: \$7,928M GAAP (tax credit benefit) / \$2,485M Non-GAAP. Highest GAAP: Q4 2023 (\$7.93B); Highest Non-GAAP: Q2 2023 (\$3.15B).	TSLA-Q4-2023.pdf (p. 5) TSLA-Q2-2023.pdf (p. 3, 5) TSLA-Q3-2023.pdf (p. 5) TSLA-Q1-2023.pdf (p. 19)
Q3	YoY Revenue Comparison <i>Comparative growth</i>	Revenue in the latest quarter (Q4 2023) grew 3% year-over-year (YoY) to \$25.2B (compared to \$24.3B in Q4 2022).	TSLA-Q4-2023.pdf (p. 4, 6) TSLA-Q3-2023.pdf (p. 5)
Q4	Management Demand & Outlook <i>Strategic synthesis</i>	Management emphasized cost reduction per vehicle, expanding operational efficiency, and aggressive investment in long-term high-potential drivers (AI compute/FSD, next-gen vehicle architecture, Cybertruck ramp, Energy Storage) amidst uncertain macroeconomic interest rate conditions.	TSLA-Q1-2023.pdf (p. 2, 28) TSLA-Q2-2023.pdf (p. 2, 3) TSLA-Q3-2023.pdf (p. 25) TSLA-Q4-2023.pdf (p. 31)
Q5	Fastest Growing Segment <i>Segment performance</i>	Energy Storage deployments grew fastest, increasing 90% YoY in Q3 2023 and reaching 125% YoY growth for full-year 2023 (14.7 GWh in 2023 vs. 6.5 GWh in 2022).	TSLA-Q4-2023.pdf (p. 9, 12) TSLA-Q3-2023.pdf (p. 7, 9) TSLA-Q1-2023.pdf (p. 7)

5. Test Benchmark Results (Continued)

#	Question & Intent	RAG System Answer	Verified Sources
Q6	Operating Margin Trend <i>Trend trajectory</i>	Operating margins across 2023: Q1: 11.4% → Q2: 9.6% → Q3: 7.6% → Q4: 8.2%. Trend was predominantly falling through Q3 due to pricing adjustments and AI/R&D; expenses, before stabilizing in Q4.	TSLA-Q1-2023.pdf (p. 5, 21) TSLA-Q2-2023.pdf (p. 5, 19) TSLA-Q3-2023.pdf (p. 5, 18) TSLA-Q4-2023.pdf (p. 31)
Q7	Dividend Declaration <i>Metric absence test</i>	'The information is not available in the uploaded documents.' <i>(Verified honest refusal — Tesla does not declare cash dividends on common stock; model strictly avoided hallucinating).</i>	TSLA-Q1-2023.pdf (p. 26) TSLA-Q2-2023.pdf (p. 24) TSLA-Q4-2023.pdf (p. 29)
Q8	Risks, Headwinds & Challenges <i>Risk factor extraction</i>	1. High interest rates & macroeconomic financing constraints. 2. EV & energy market competitive pricing pressure. 3. 4680 battery cell scaling & Cybertruck manufacturing ramp. 4. Foreign exchange fluctuations & regulatory incentive policy shifts.	TSLA-Q1-2023.pdf (p. 28) TSLA-Q2-2023.pdf (p. 26) TSLA-Q3-2023.pdf (p. 25) TSLA-Q4-2023.pdf (p. 31)
Q9	3-Line Client Email Summary <i>Executive synthesis</i>	• Revenue & Deliveries: Tesla reached \$25.2B Q4 revenue (+3% YoY), capping record 1.81M full-year deliveries. • Profitability: Operating margin stabilized at 8.2% with \$2.1B operating income via tight cost controls. • Growth Drivers: Energy Storage surged (+125% YoY), while investments centered on AI compute and next-gen platforms.	TSLA-Q4-2023.pdf (p. 1, 4, 6) TSLA-Q3-2023.pdf (p. 1, 5)
Q10	CEO 2015 Shareholding <i>Adversarial trap question</i>	'The information is not available in the uploaded documents.' <i>(Passed adversarial test — knowledge base contains only FY23 data; system correctly declined to fabricate 2015 data).</i>	TSLA-Q1-2023.pdf (p. 26) TSLA-Q4-2023.pdf (p. 28)

6. Engineering Insights & Challenges ("What Did Not Work Well")

- **Complex Multi-Column Table Layouts:** Earnings decks frequently place condensed income statements and balance sheets into side-by-side graphical columns. Standard character chunkers occasionally detach column headers from numerical rows. Fine-tuning chunk size to 1000 characters with 200-character overlap preserved numerical coherence.
- **Cross-Quarter Retrieval Calibration:** Questions comparing trends across all four quarters require retrieving chunks from four independent PDF documents simultaneously. Standard top_k=4 was insufficient; calibrating retrieval to top_k = 8 to 20 ensured all four quarterly summary tables entered the prompt context window.
- **Streaming Adapter Design:** Streamlit leverages Python generator streams (`st.write_stream()`) for micro-latency UX, whereas FastAPI REST endpoints require resolving generators into unified JSON strings before serialization.

7. Project Resources & Quick Access

GitHub Repository:	https://github.com/SreeAditya-Dev/Finance-RAG
Video Demo Walkthrough:	https://res.cloudinary.com/dukitmdtn/video/upload/v1786784336/Screen_Recording_2026-08-15_142015_wsvg6i.mp4
Local FastAPI Documentation:	http://localhost:8000/docs POST /ask, POST /ingest, GET /stats
Local Streamlit Application:	http://localhost:8501 Interactive UI with source verification badges